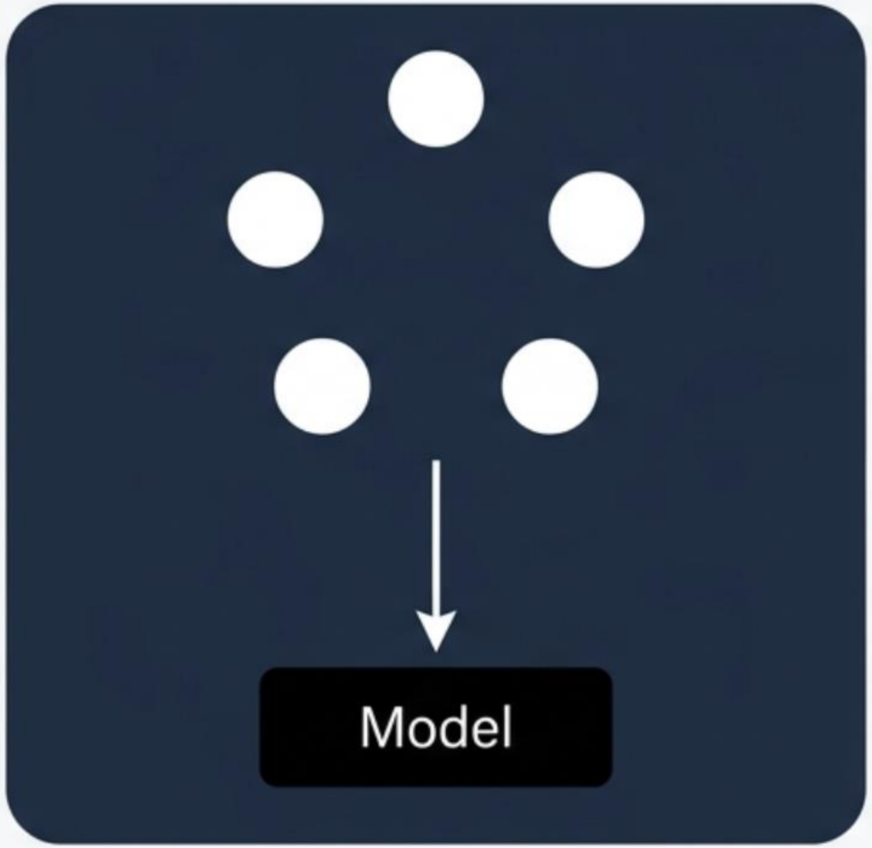
EVALUATING THE IMPACT OF DISCOVERED CAUSAL STRUCTURES ON STRUCTURE-AWARE SHAPLEY METHOD

An Experimental Diagnostic for ML practitioners

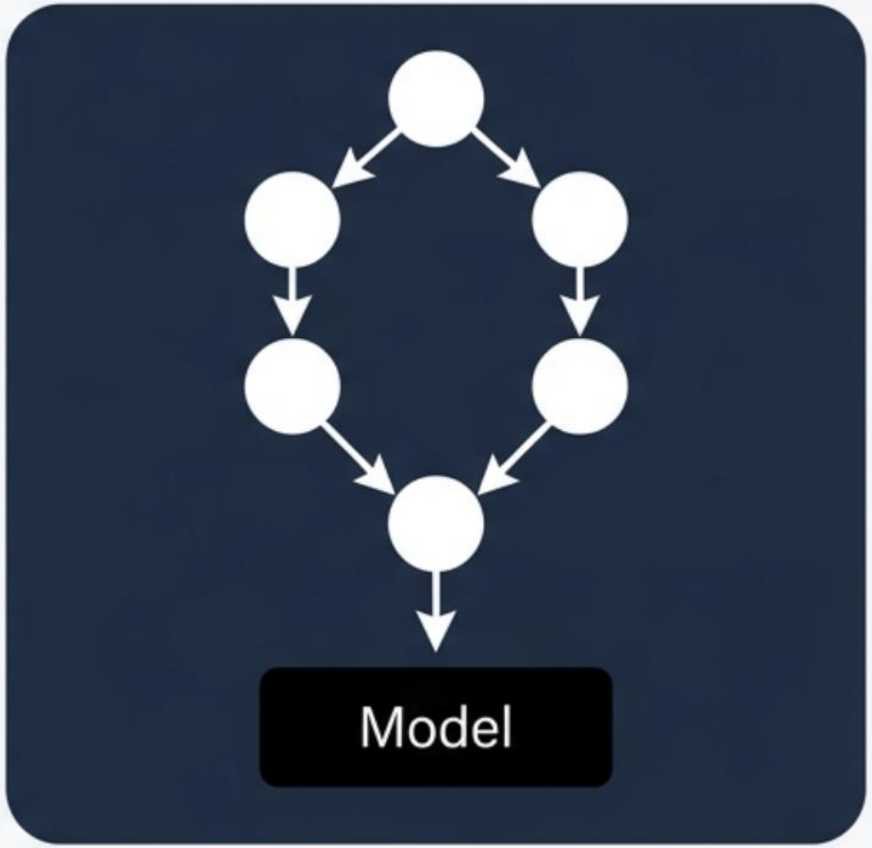
Juan David Rios

The Independence Assumption



Traditional Shapley treats every feature as an independent, interchangeable coalition partner.

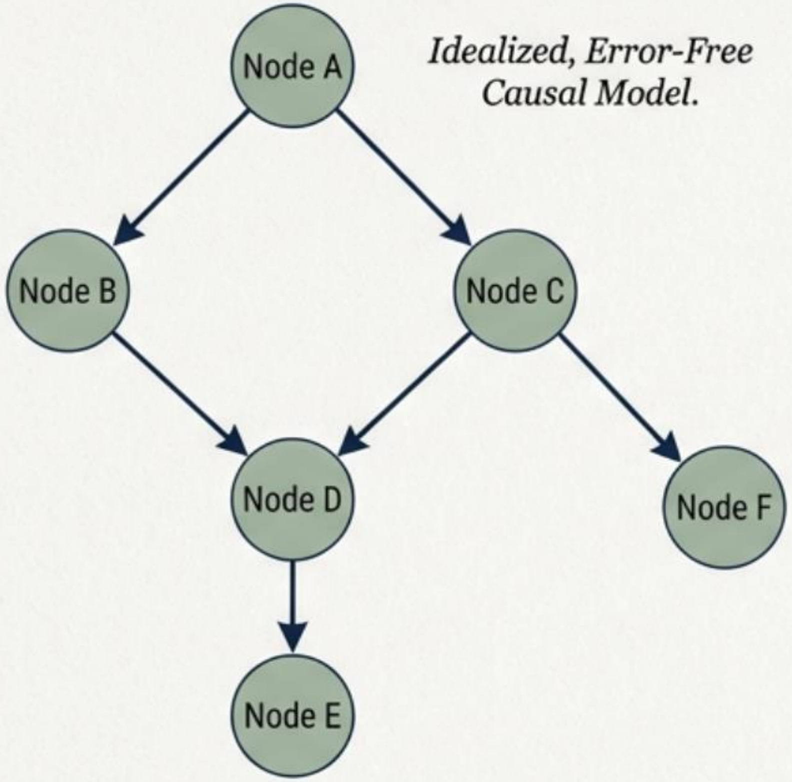
The Causal Reality



Real-world generative processes involve structured dependencies. Treating downstream effects and upstream causes equally misattributes credit.

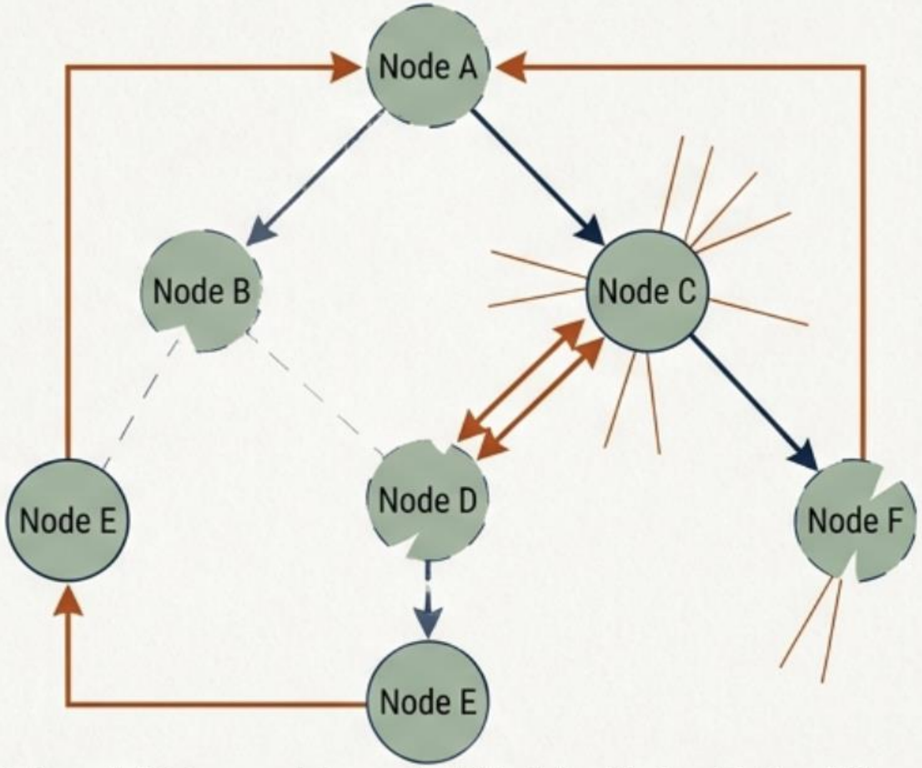
THE LITERATURE'S ASSUMPTION

A validated, ground-truth causal graph exists to to map feature dependencies perfectly.



THE PRACTITIONER'S REALITY

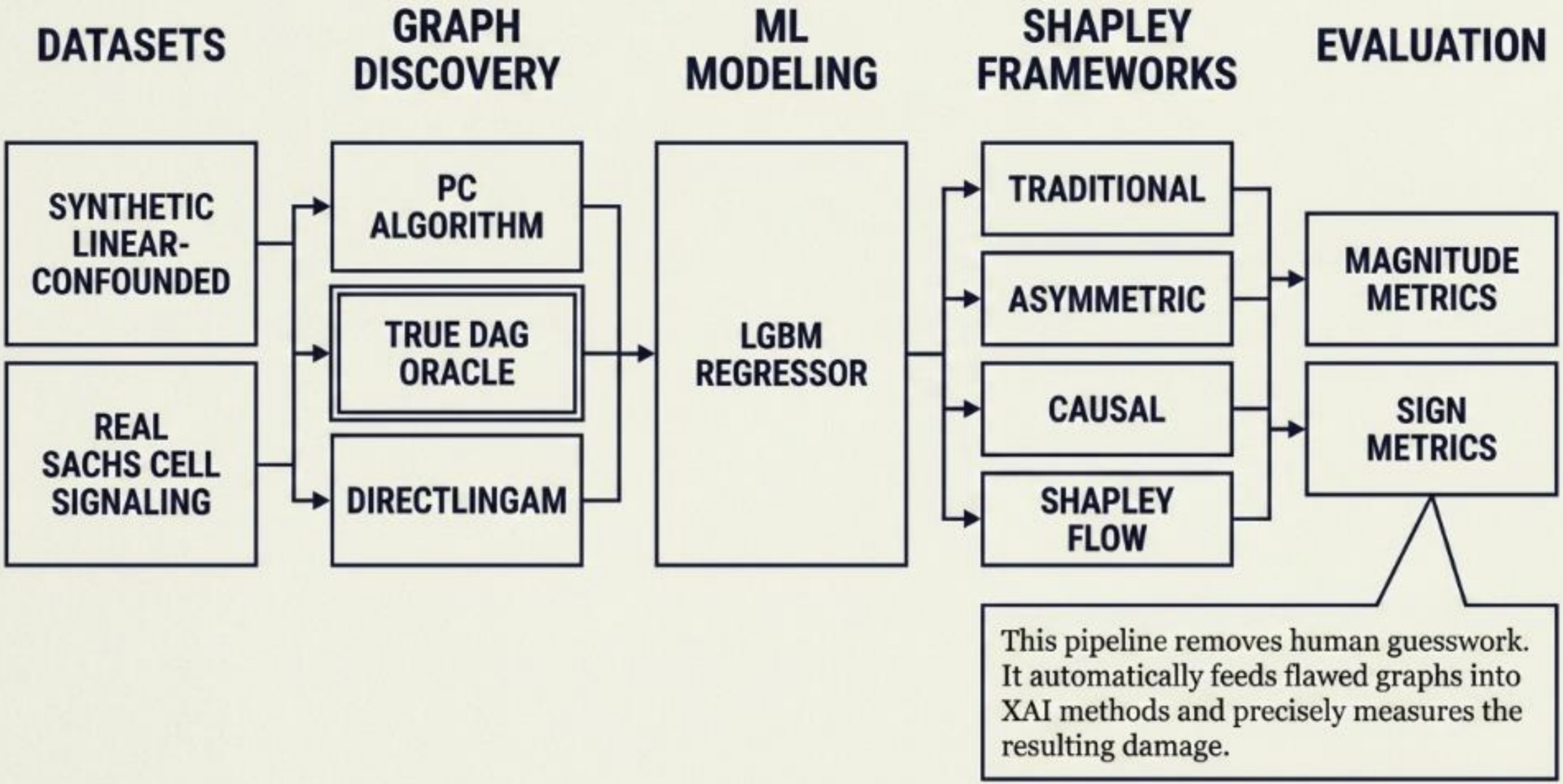
We rely on automated discovery algorithms. The inferred graphs are noisy, imperfect, and contested.



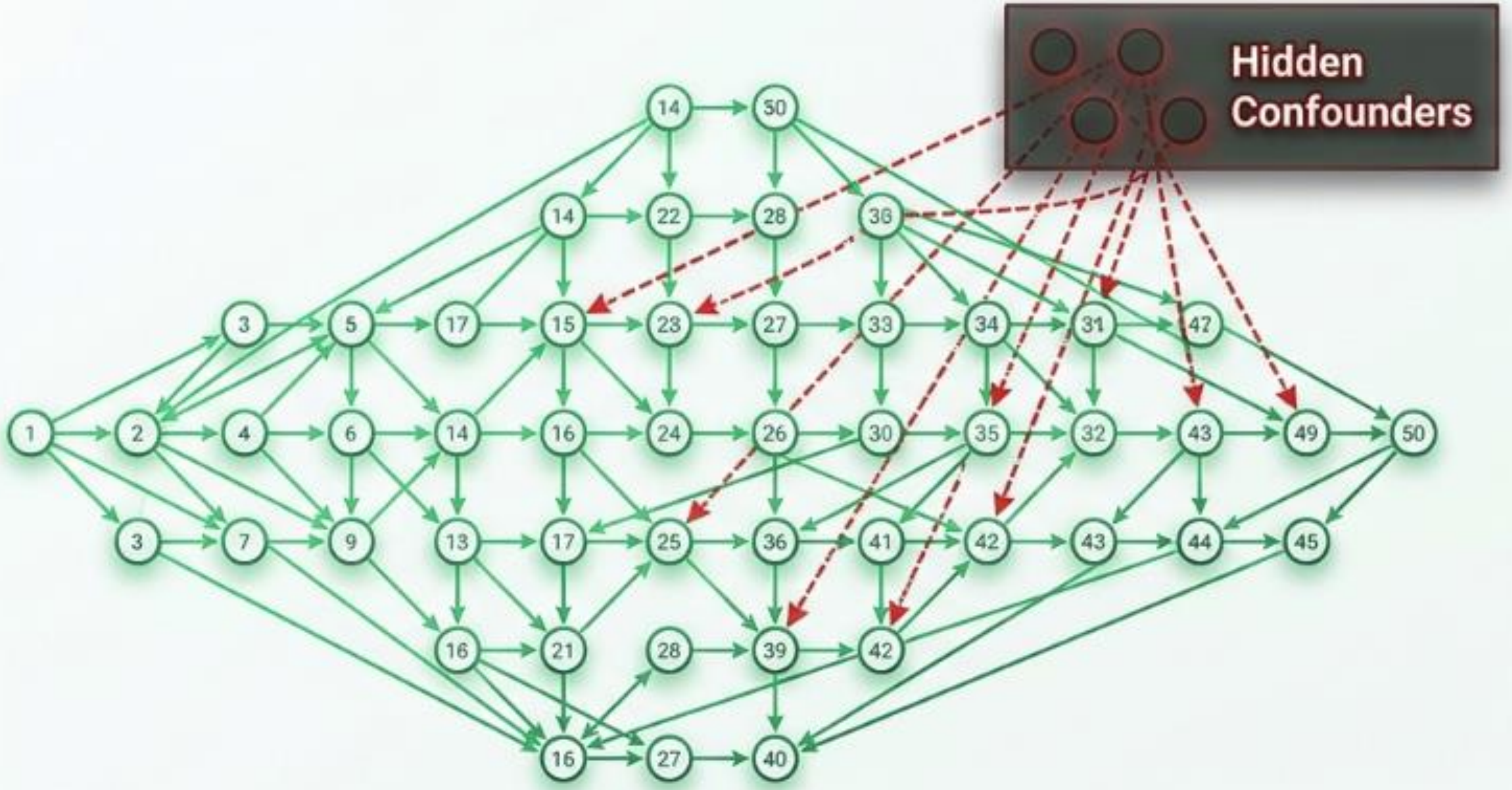
Inferred, Noisy, and Imperfect Model with Conflicting Edges.

STRUCTURE-AWARE EXPLAINABLE AI (XAI) ASSUMES THE CAUSAL MAP IS CORRECT.

What happens to our explanations when the map is wrong?


DATASETS
GRAPH DISCOVERY
ML MODELING
SHAPLEY FRAMEWORKS
EVALUATION
SYNTHETIC LINEAR-CONFOUNDED
REAL SACHS CELL SIGNALING
PC ALGORITHM
TRUE DAG ORACLE
DIRECTLINGAM
LGBM REGRESSOR
TRADITIONAL
ASYMMETRIC
CAUSAL
SHAPLEY FLOW
MAGNITUDE METRICS
SIGN METRICS
This pipeline removes human guesswork. It automatically feeds flawed graphs into XAI methods and precisely measures the resulting damage.
NotebookLM

Setting the Baseline with a Controlled Synthetic Stress Test



System

Purely linear structural equations with 1,000 samples.

Scale

50 independent features.

The Stressor

Causal Sufficiency is deliberately violated by 5 hidden confounders, simulating unobserved real-world variables.

The Real-World Benchmark in Biological Cell Signaling

Dataset

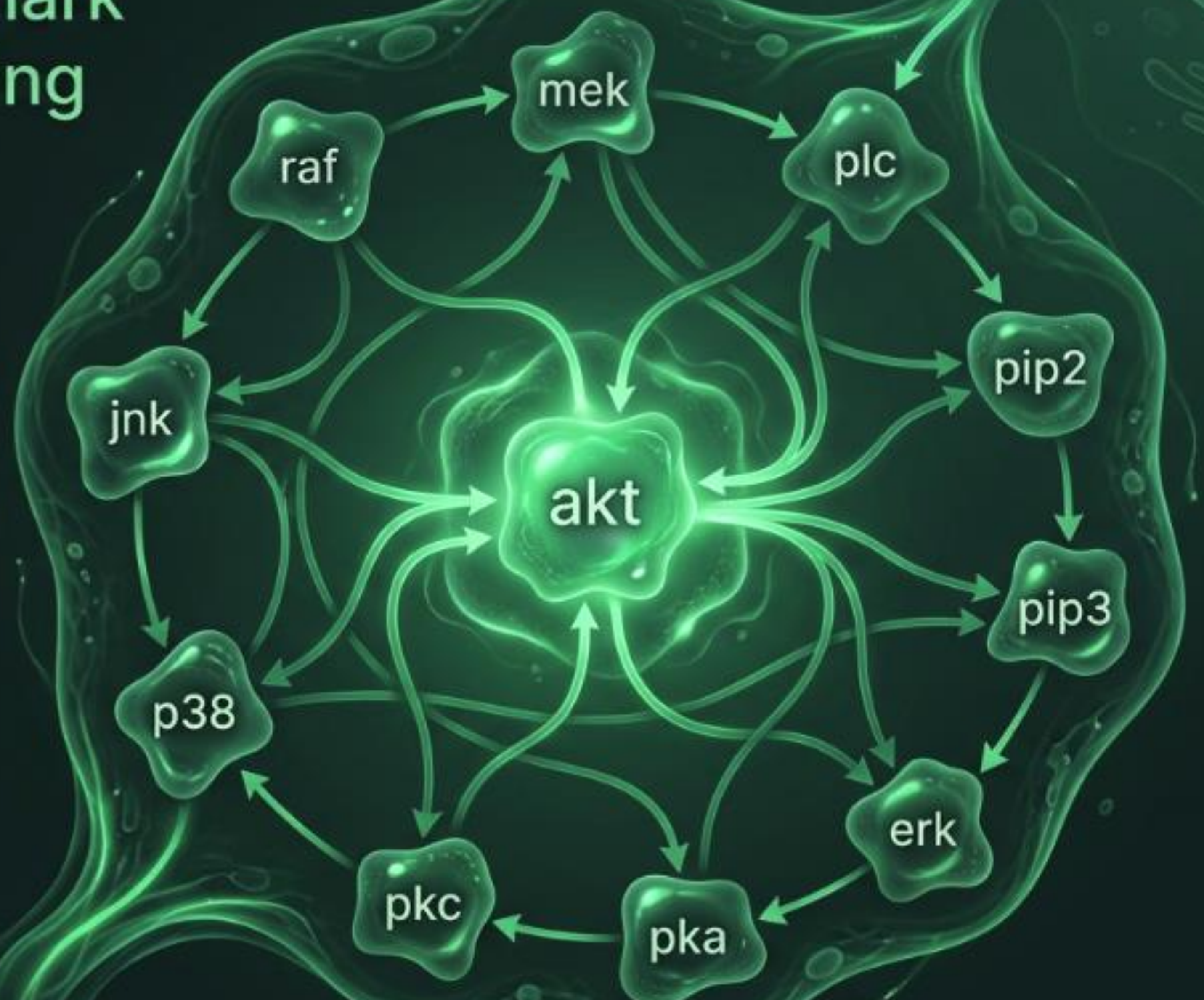
The gold-standard Sachs cell signaling dataset. 7,466 simultaneous measurements of human T-cells.

The Target

Predicting the activity level of the Akt kinase.

The Ground Truth

Unlike typical real-world data, this system has a biologically validated consensus DAG (17 edges) serving as an objective reference.

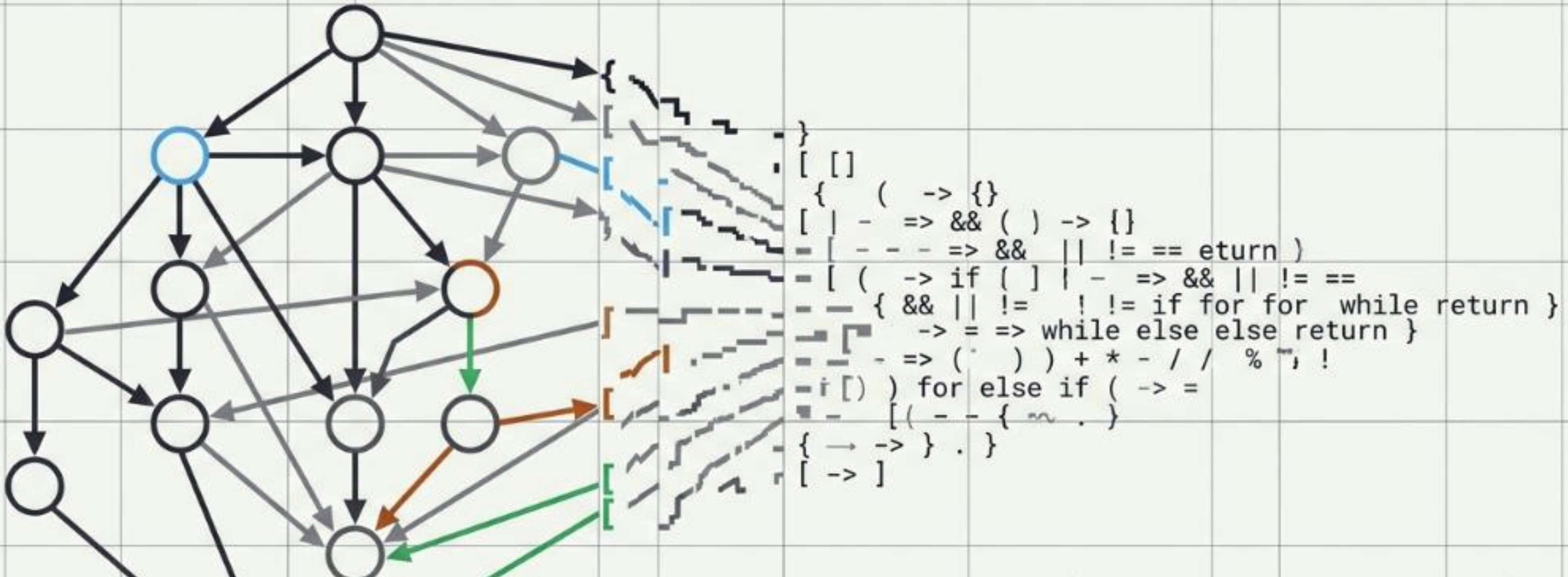


The Causal Discovery Irony Separating Prediction from Reality

The Expectation: We deploy highly accurate predictive models...

...but attempt to explain them using fundamentally broken causal graphs.

If our causal graphs are this noisy, what happens when we inject them into our explainability algorithms?



The Architecture of Causal Explanations

Theoretical Foundations and Computational Execution of Structure-Aware Shapley Methods

The Causal Shapley Matrix

How structure-aware methods rewrite the rules of the cooperative game.

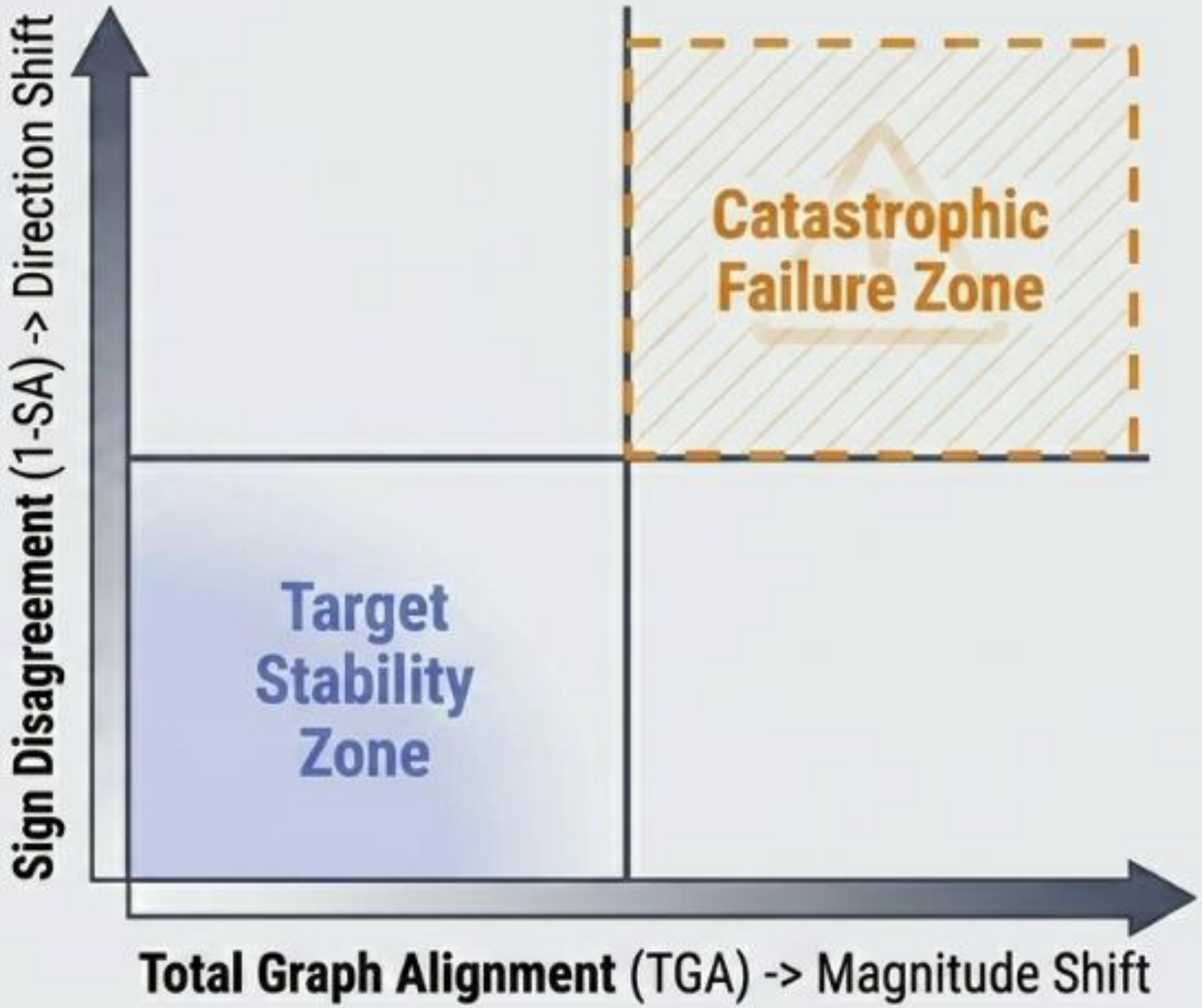
	Traditional Shapley	Asymmetric Shapley	Causal Shapley	Shapley Flow
Players	Features	Features	Features	**Directed Edges**
Permutation Space	All $N!$ orderings	**Topological orderings of the feature DAG**	Topological orderings of the component DAG	All orderings of the edge set
Value Function	Observational $\mathbb{E}[f \mid X_S = x_S]$	Observational $\mathbb{E}[f \mid X_S = x_S]$	**Interventional** $\mathbb{E}[f \mid \mathrm{do}(X_S = x_S)]$	**Foreground/ background by edge activation**
Graph's Role	None	**Constrains the ordering**	**Defines components, ordering, and intervention propagation**	**Supplies the players (edges)**

Evaluating Failure Across Two Independent Axes

Most explainability research only measures magnitude.

But a sign reversal is a **catastrophic failure** for a practitioner: a graph that inverts signs tells the exact opposite story to the user.

A **stable method** must sit in the bottom-left corner of this matrix.

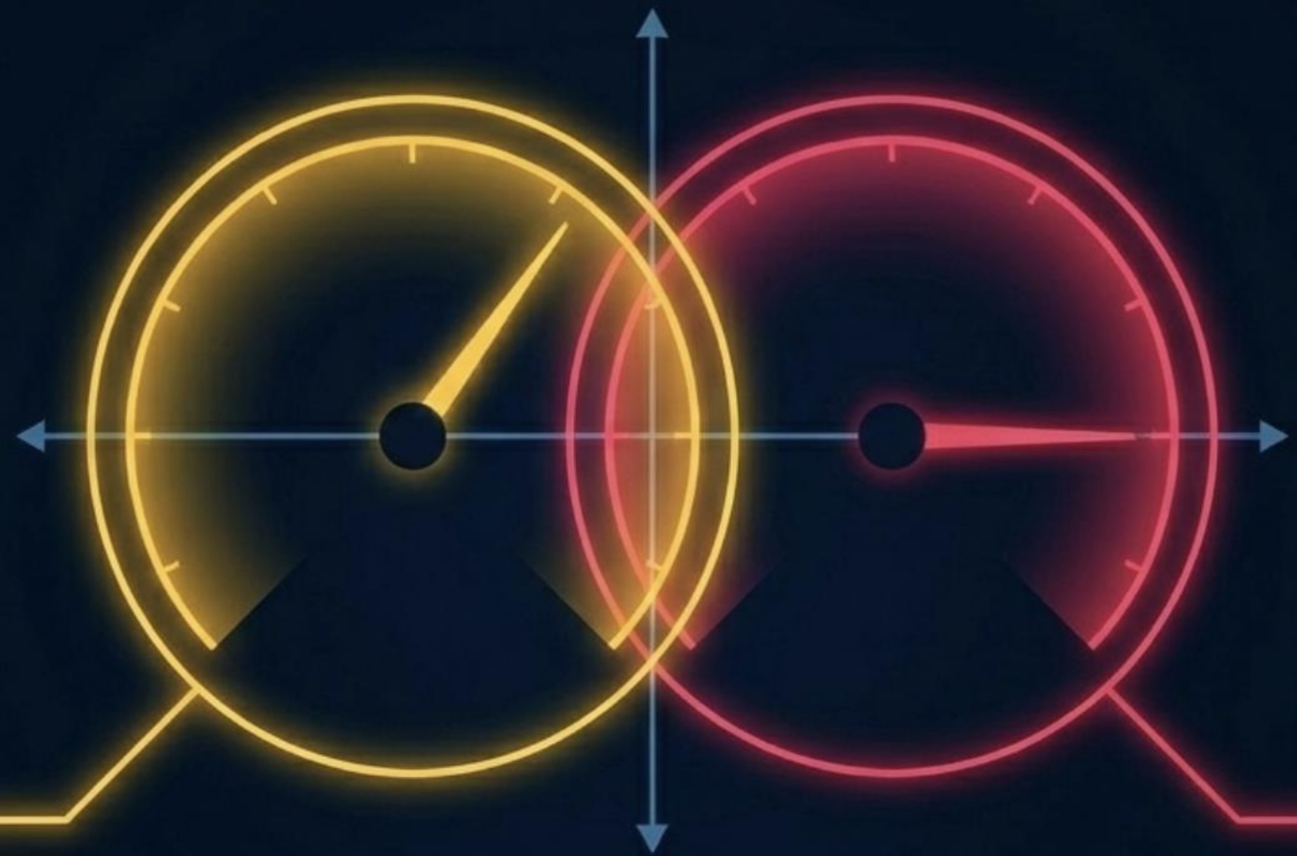


Core Diagnostic Framework

Magnitude Divergence (ΔM)

Measures how much the absolute size of an attribution diverges, expressed as a fraction of the model-output standard deviation.

Meaning: Rescaling. The graph alters the weight of the feature, but preserves the qualitative story.



Sign Disagreement (D)

The percentage of instances where the feature attribution points in the opposite direction to the reference.

Meaning: Inversion. The graph tells the affected individual the exact opposite story of why a decision was made.

A magnitude shift rarely crosses zero; a sign flip happens at the zero boundary. To evaluate structural fragility, both must be measured simultaneously.

Explainable Zoom Baselines

Three Evaluation Baselines

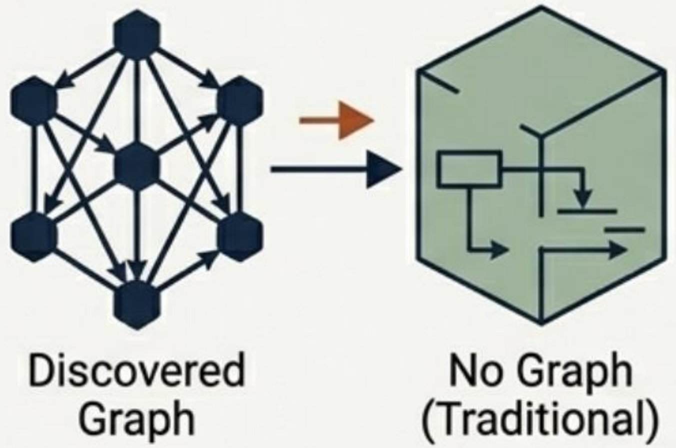
- **vs. Baseline**: How far does injecting a graph move us from Traditional Shapley?
- **vs. Cross-Discovery**: How much does the explanation change if we swap PC for LiNGAM?
- **vs. Oracle**: How faithfully does the method recover the True DAG?

Navigating Without a Ground-Truth Map

The Baseline Reference

Discovered Graph vs. No Graph (Traditional)

Question: How much does injecting any structural theory perturb the standard explanation?

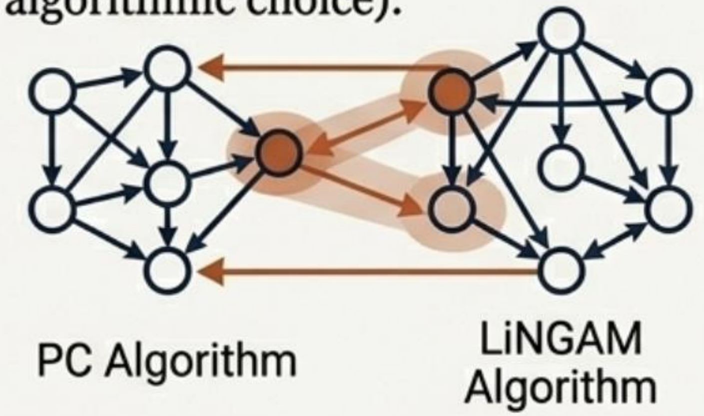


Question: How much does injecting any structural theory perturb the standard explanation?

The Cross-Discovery Reference

PC Algorithm vs. LiNGAM Algorithm

Question: If two algorithms look at the same data, how wildly do their resulting explanations disagree? (Measures the instability of algorithmic choice).

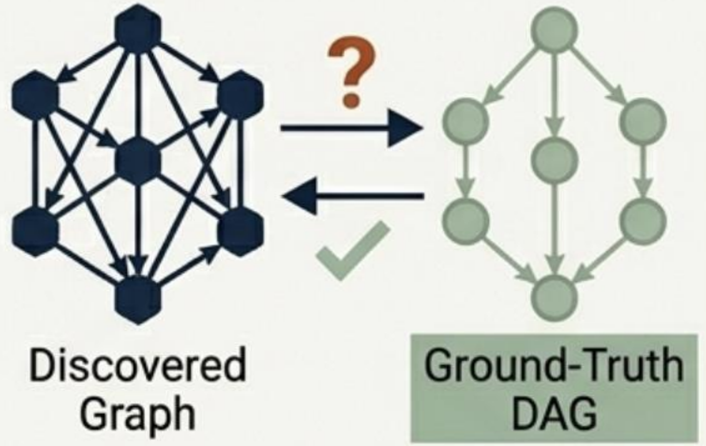


Question: If two algorithms look at the same data, how wildly do their resulting explanations disagree? (Measures the instability of algorithmic choice).

The Oracle Reference

Discovered Graph vs. Ground-Truth DAG

Question: Did the graph injection actually move us closer to the truth? (Academic benchmark only).



Question: Did the graph injection actually move us closer to the truth? (Academic benchmark only).

THE TWO-SIGNAL ALARM

HIGH BASELINE DEVIATION (ΔM_{base})

Injecting any graph dramatically alters the feature's attribution compared to graph-free Traditional Shapley.

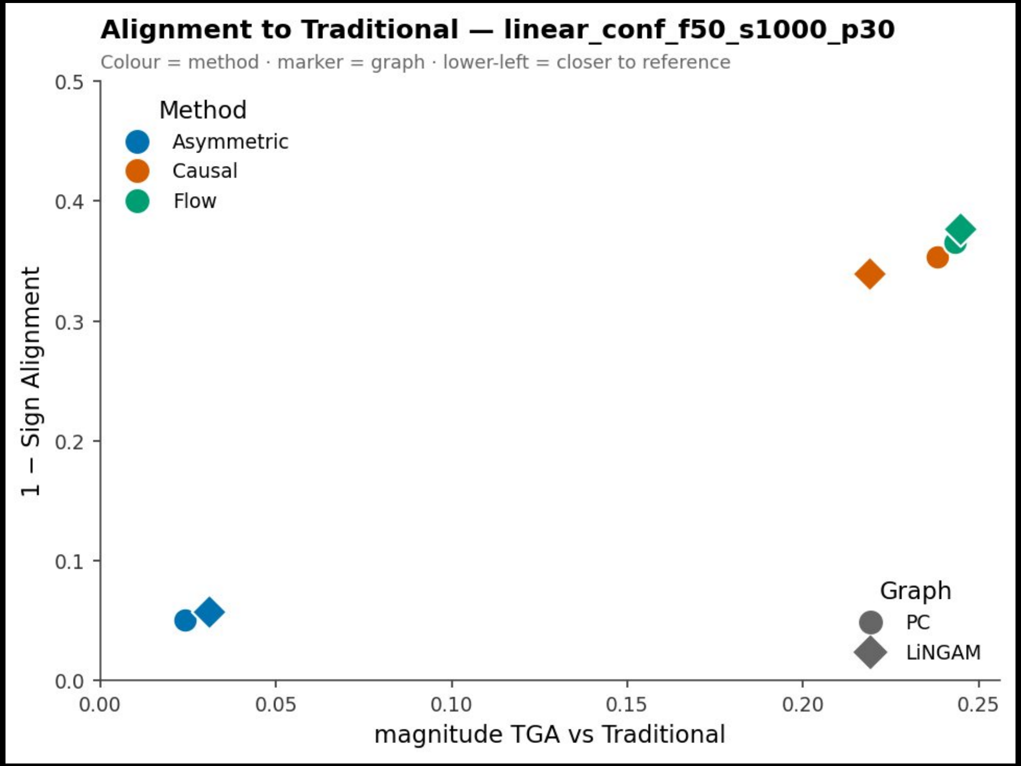
HIGH CROSS-DISCOVERY INSTABILITY (ΔM_{disc})

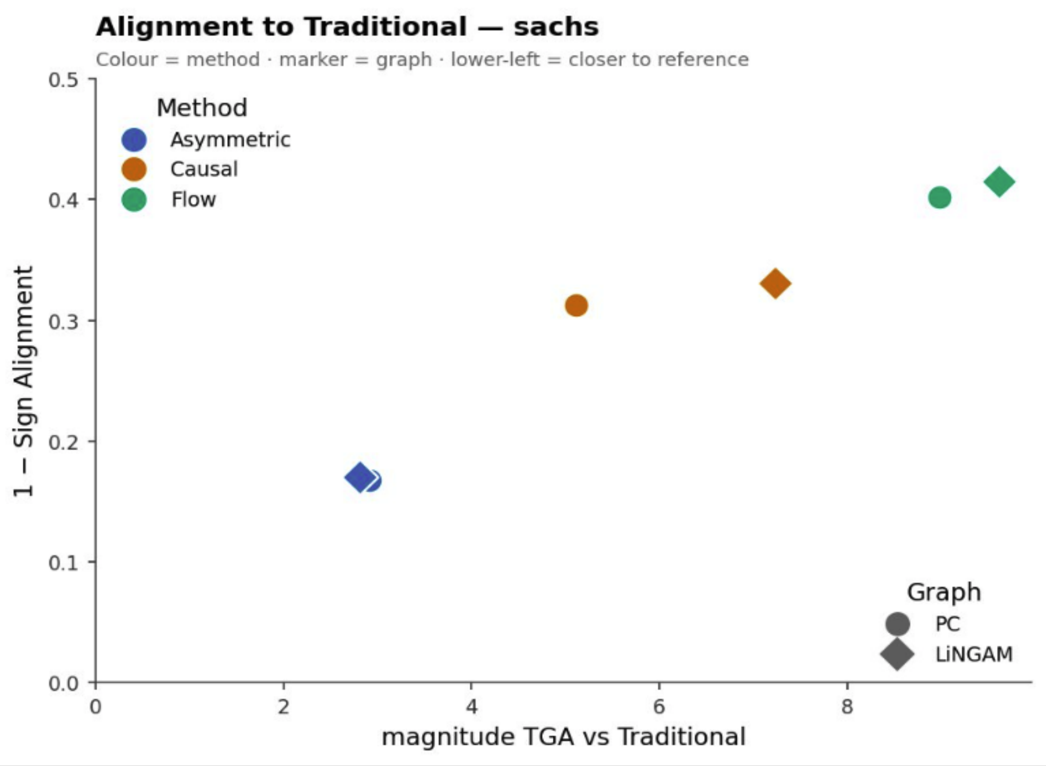
The PC algorithm and LiNGAM algorithm fundamentally disagree with each other on the attribution size.

VERDICT: When both alarms trigger on a single feature, its causal neighborhood is both highly influential AND heavily contested. This is an immediate red flag demanding manual human validation.

BASELINE DEVIATION: ADDING STRUCTURE WILDLY SHIFTS CAUSAL AND FLOW

Causal and Flow flips the direction of over one-thirds of all explanations, drastically re-weigthing the model explainability story.
Assymetrics maintains model explainabilty story, low change of magnitude, low change of direction.
There is no advantage of a discovery model having a higer F1 score or overall accuracy metrics, the node level specifics determine the changes.





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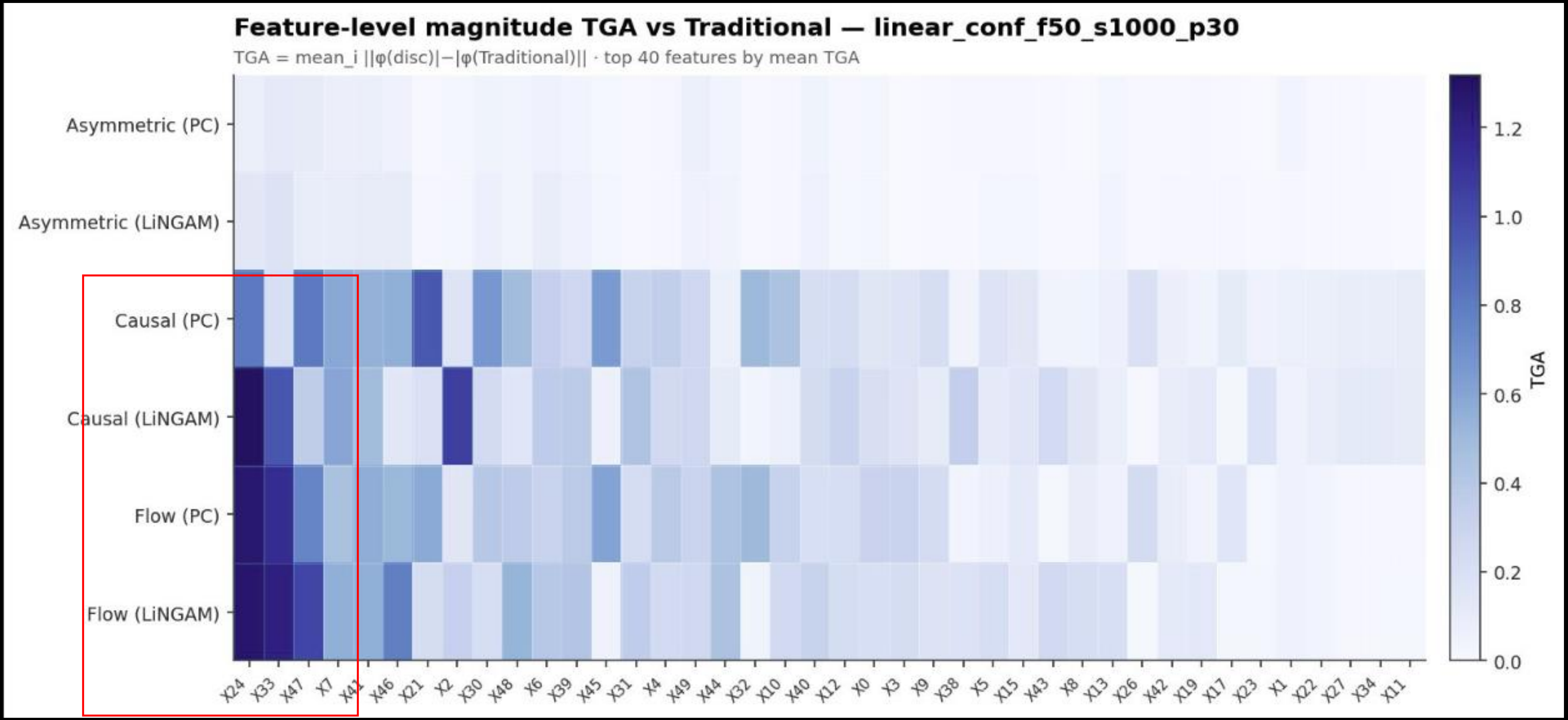

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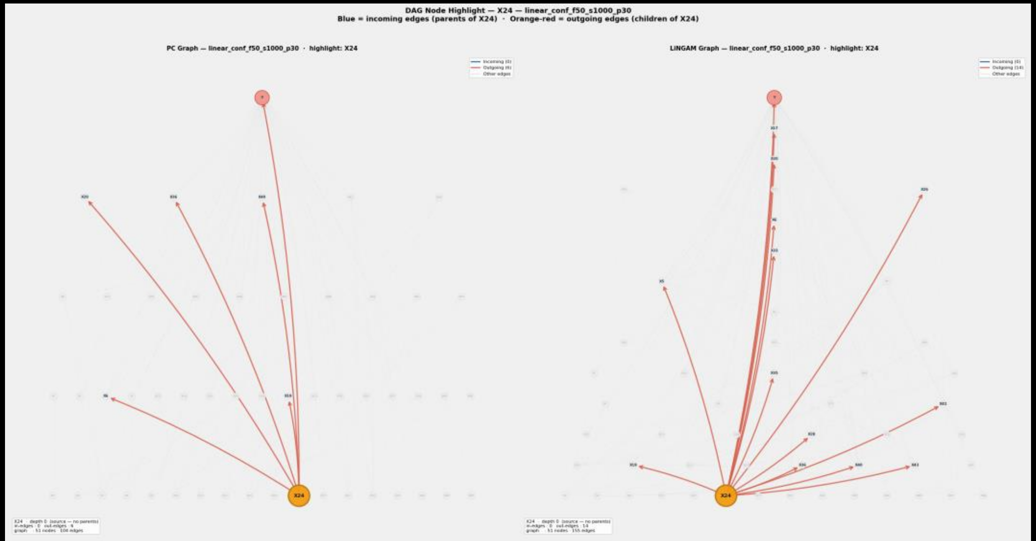
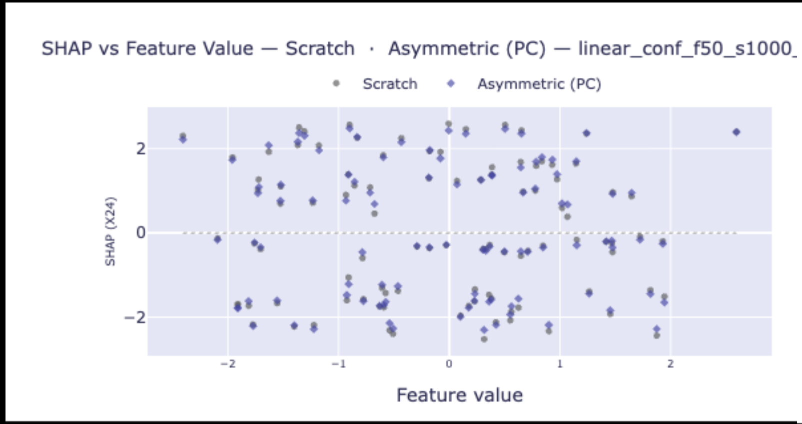
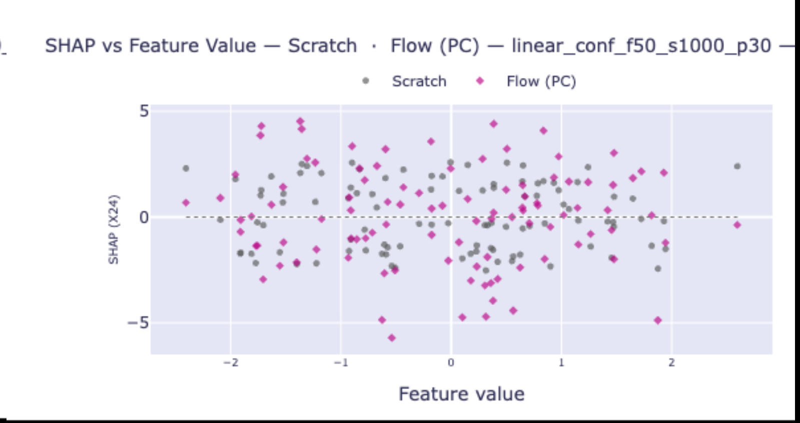
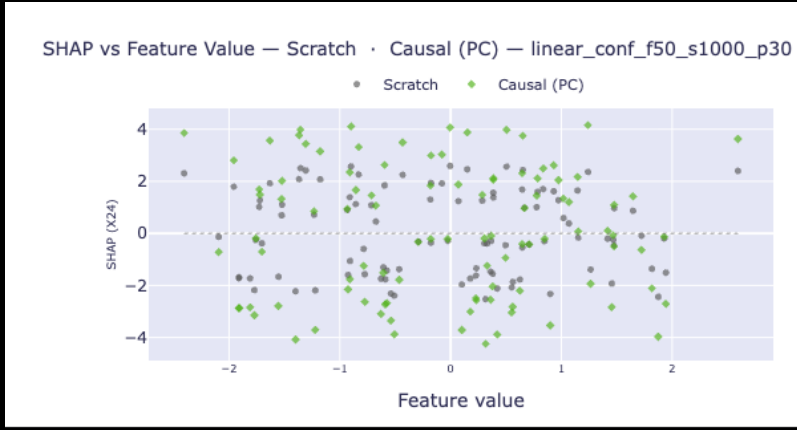


The interventional process of Causal Shapley and the edge attribution process of Shaply flow, amplified 4 times the changes for features that are Predicted as source nodes







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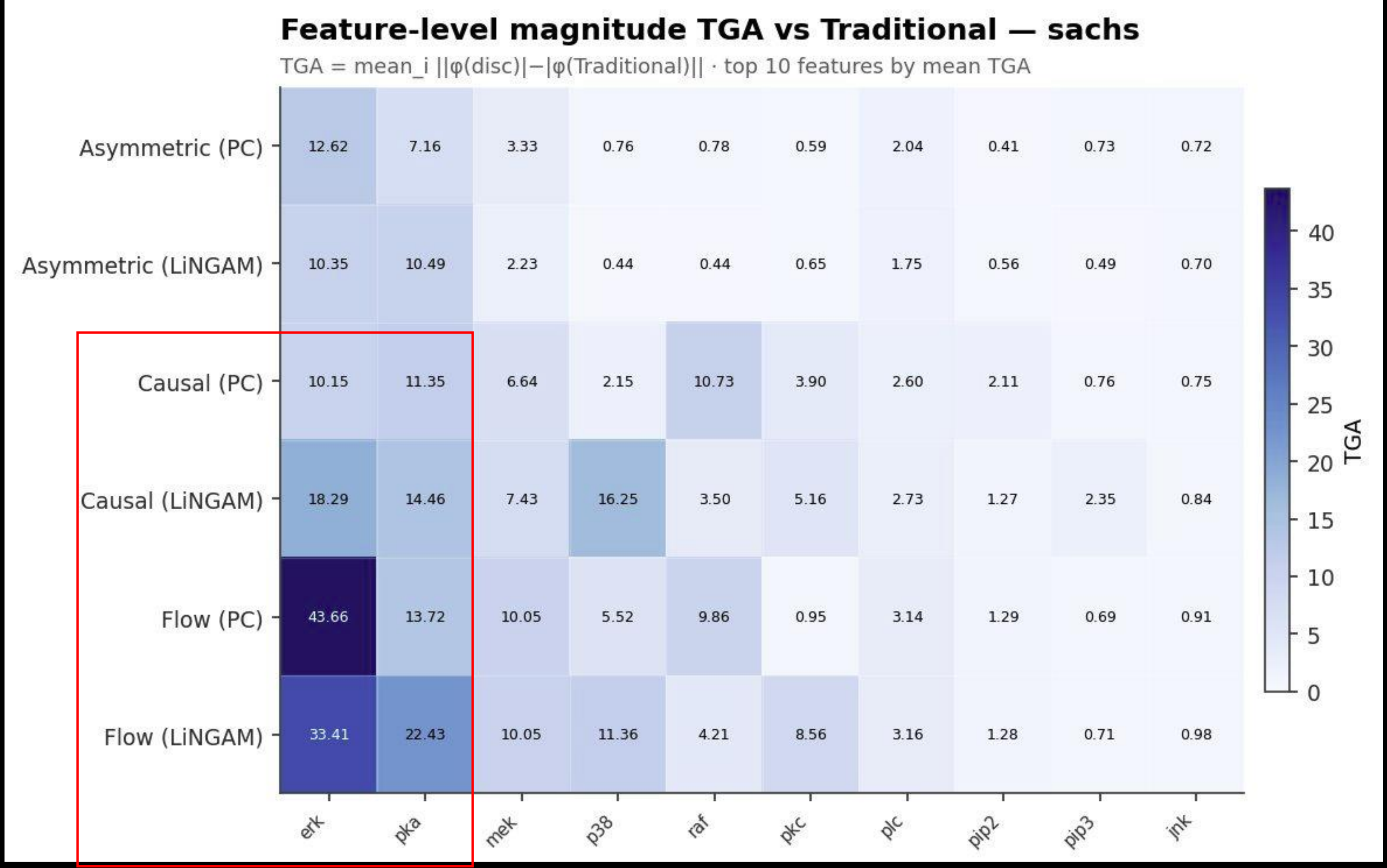

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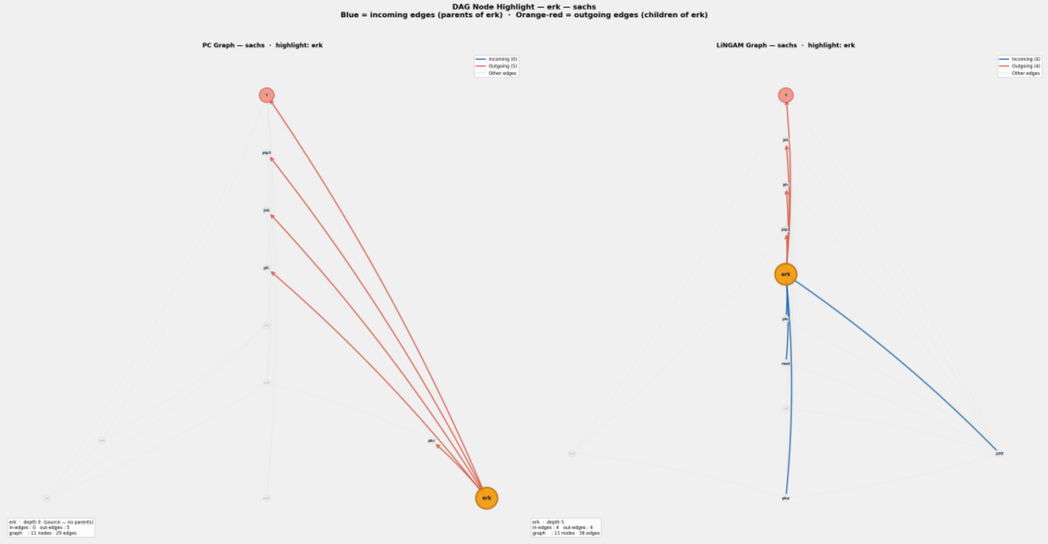
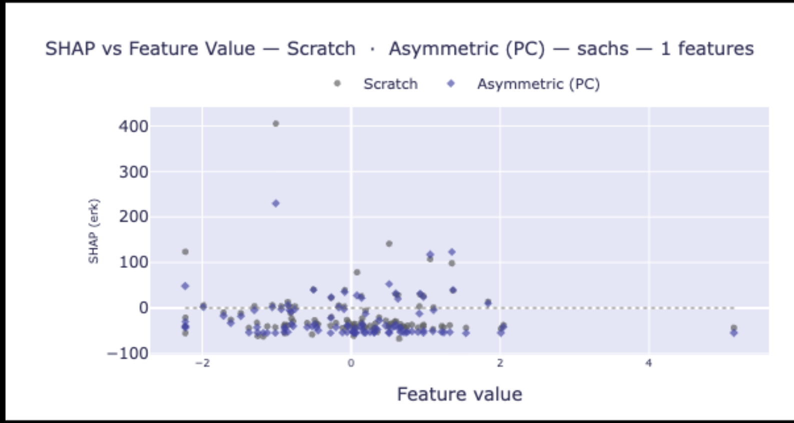
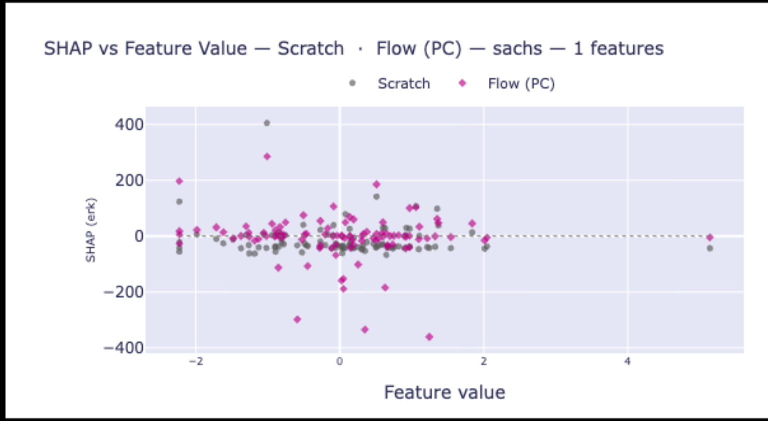
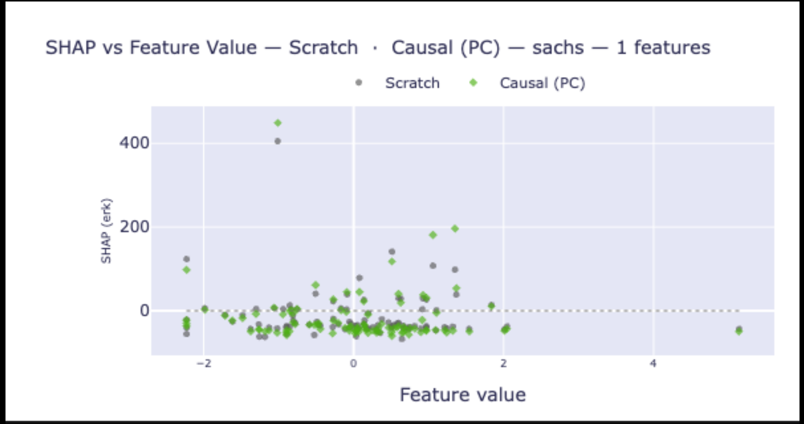


The interventional process of Causal Shapley and the edge attribution process of Shaply flow, amplified 3 times the changes for features that are Predicted as source nodes

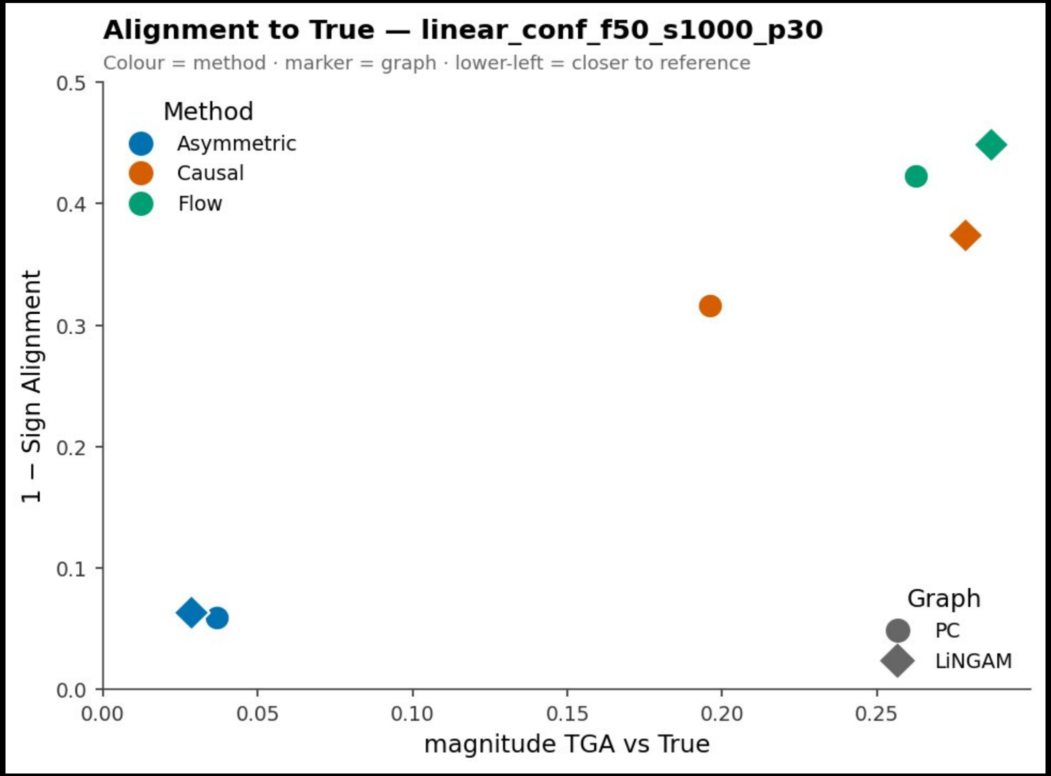


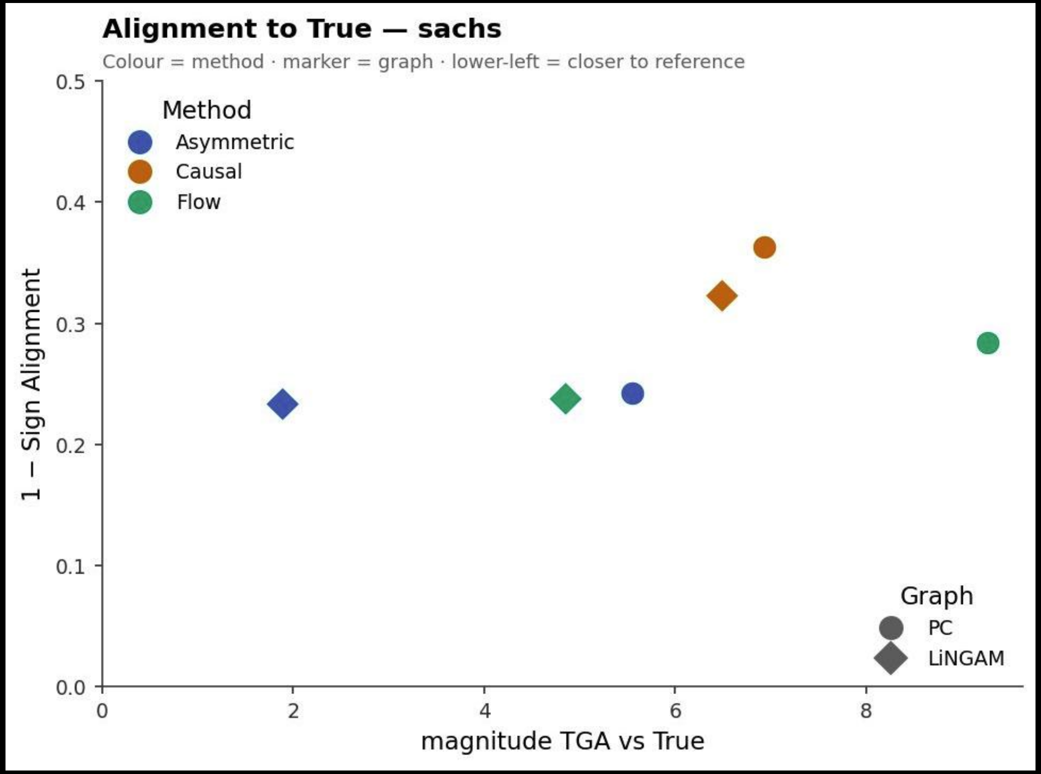




IMPERFECT GRAPH DEVIATION: THE OVERALL PRECISION MATTERS

Causal and Flow do not shift toward the truth, they shift toward the noise. Even with good overall discovery metrics, singular false predicted edges severely distort their attributions.
In the Sachs and Synthetic dataset there is an impact of having a better accuracy on the discovered graph method, we see that all shapley methods using LiNGAM have a widely lower magnitude error vs the true DAG, that is mostly that difference is more clear in the true dataset





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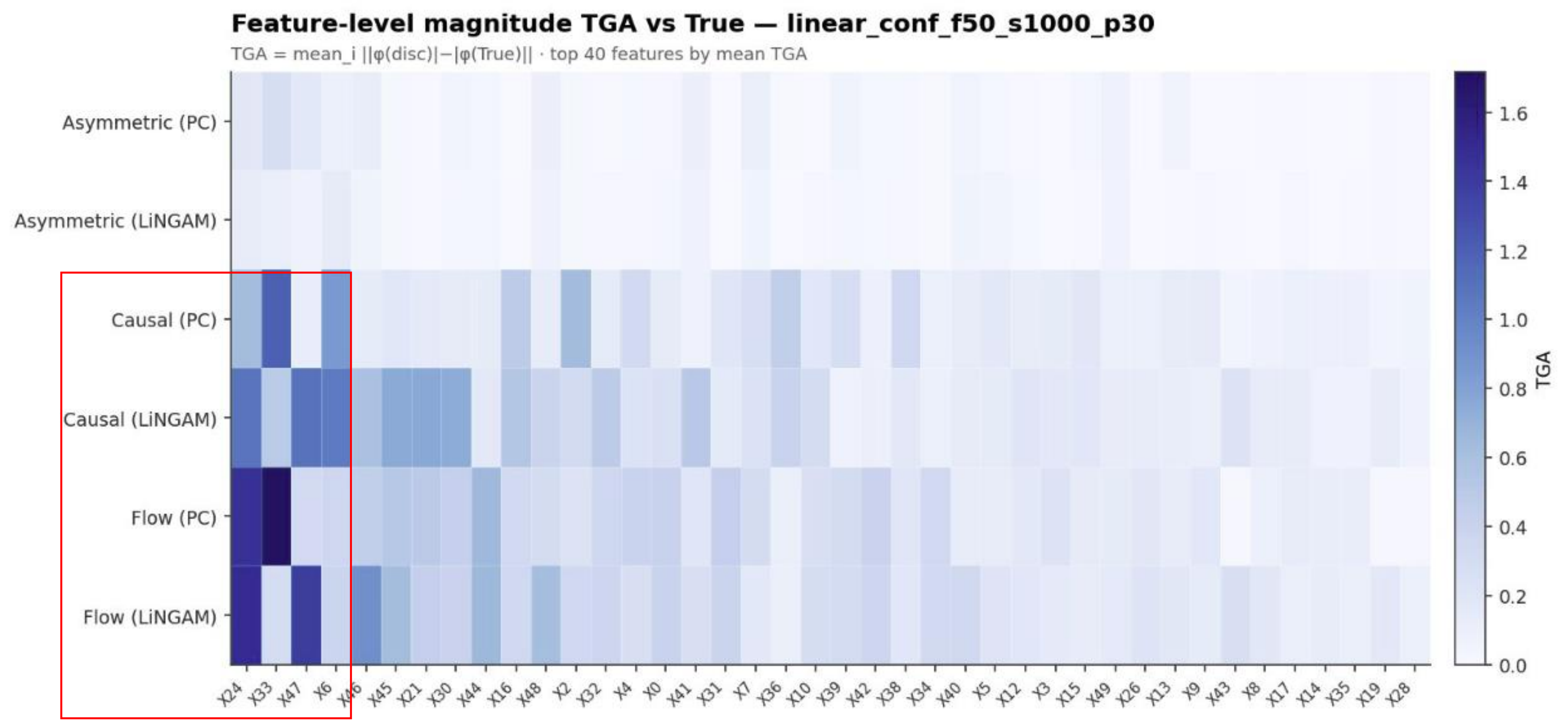
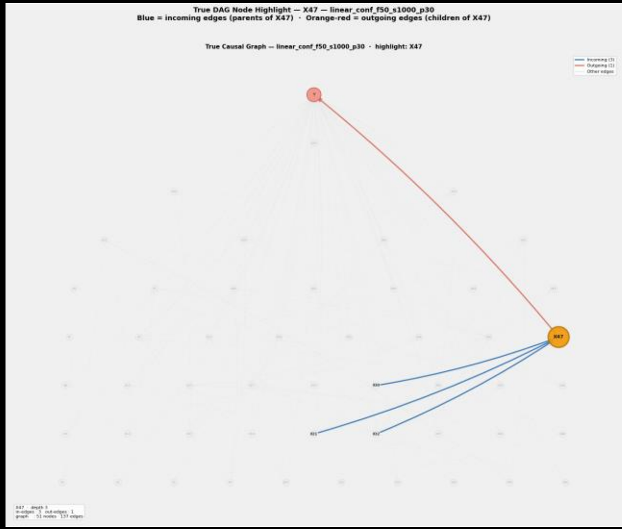
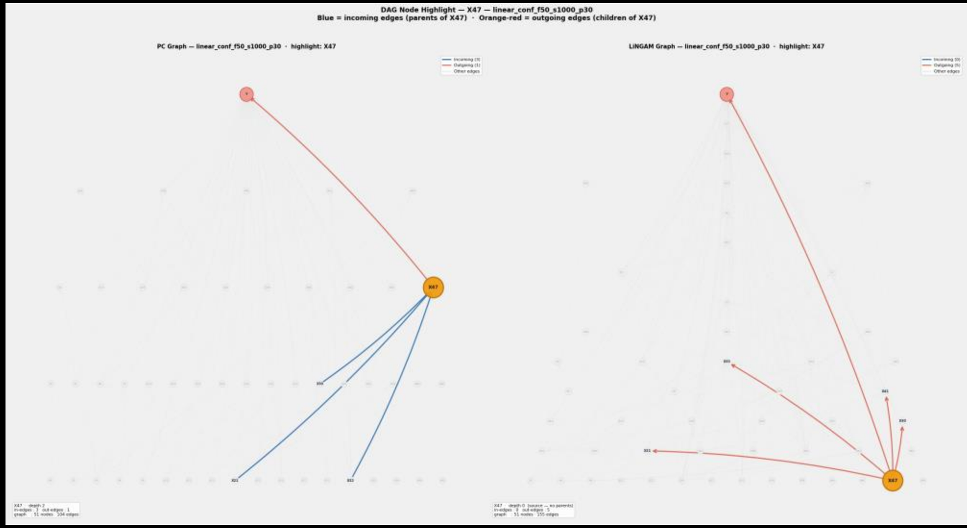
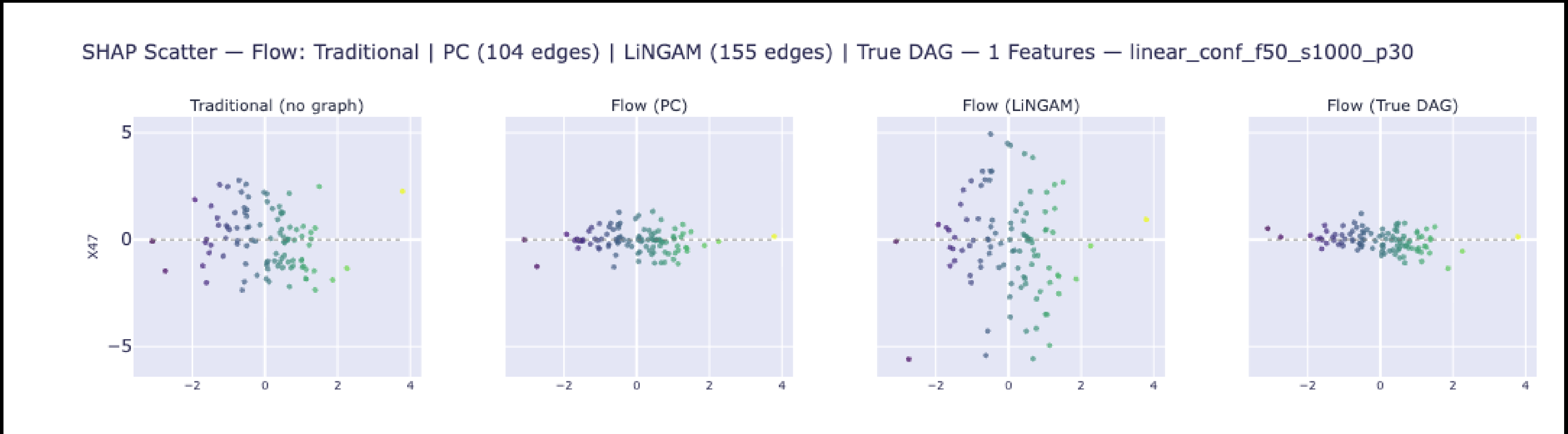


BASELINE DEVIATION: ADDINGSTRUCTURE WILDLY SHIFTS CAUSAL AND FLOW



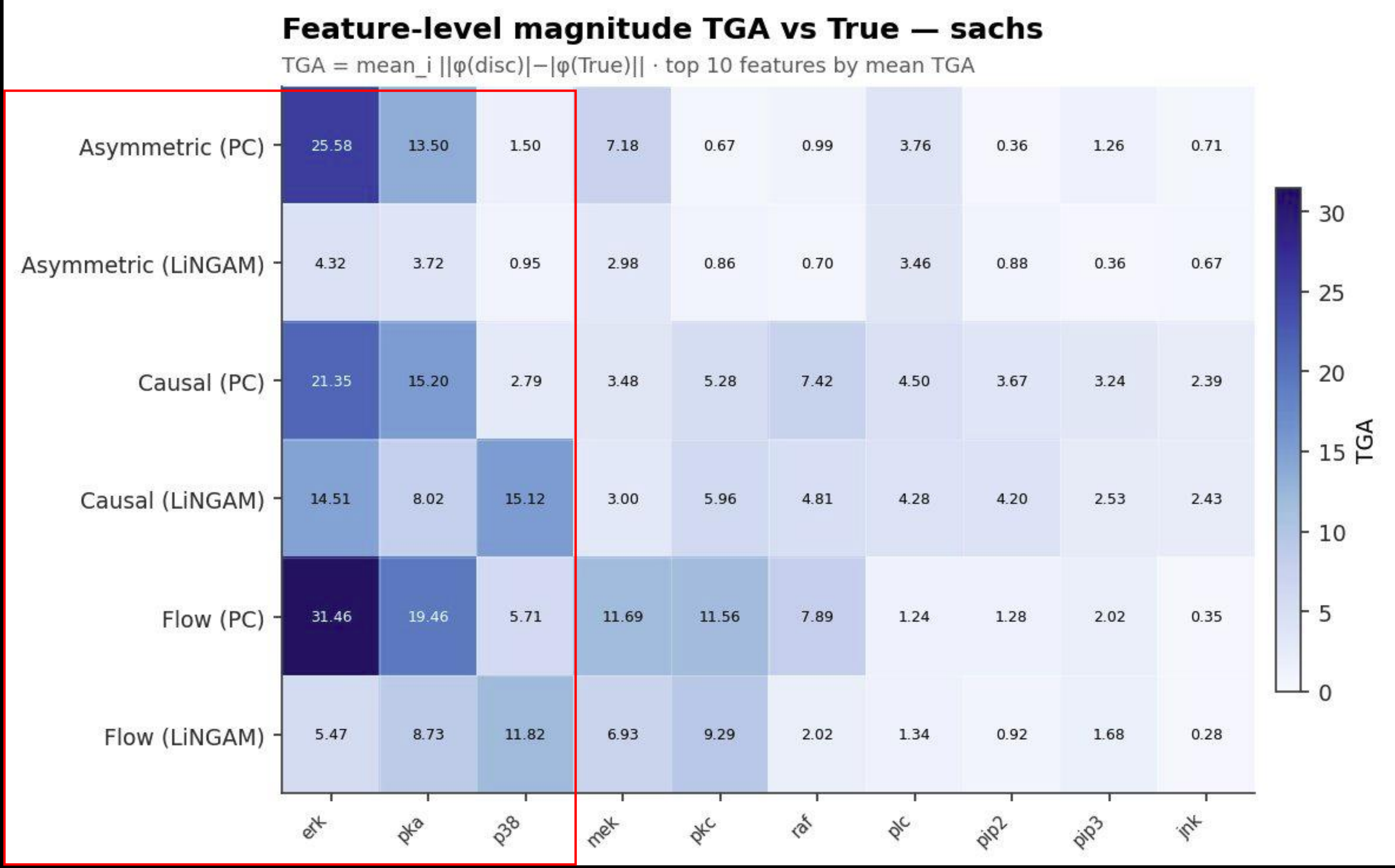




IMPERFECT GRAPH DEVIATION: THE OVERALL PRECISION MATTERS

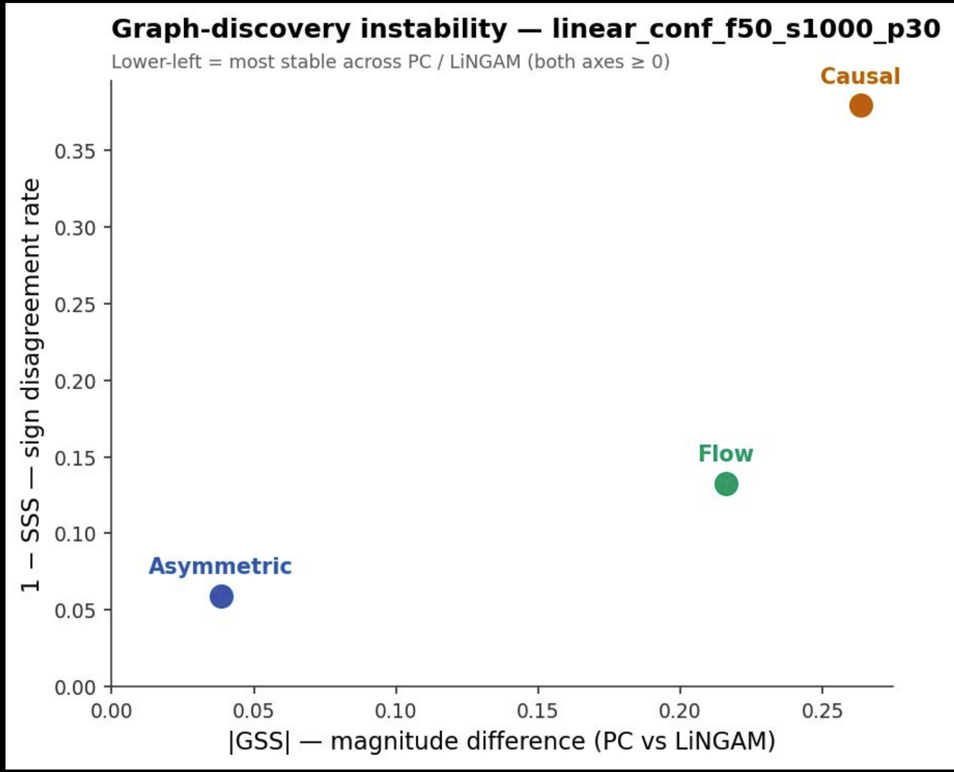
The LiNGAM Advantage:
- Because LiNGAM achieves higher causal discovery accuracy than the PC algorithm, its resulting Shapley values are significantly closer to the ground truth.

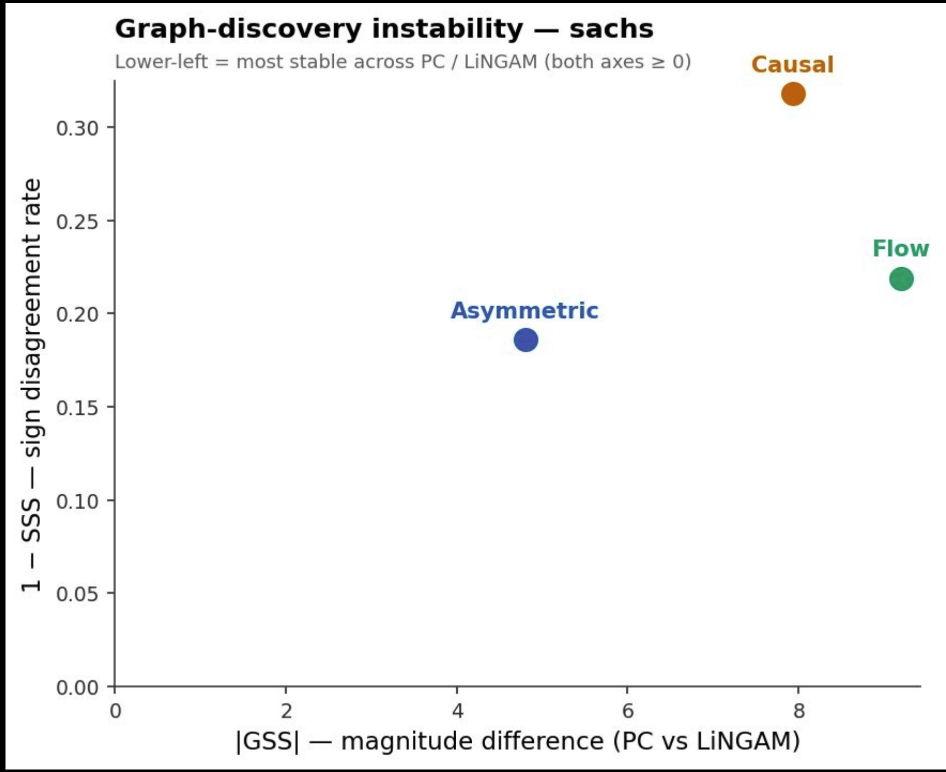
Structural Vulnerability:
- However, even with highly accurate discovery, specific graph topologies (the exact arrangement of connections) can cause attribution values to diverge sharply.



THE INSTABILITY FACTOR: DISCOVER ALGORITHM CHOICE DICTATES THE EXPLANATION

For Causal Shapley and Shapley Flow the choice between PC and LINGAM dictates the final explanation more than the model's actual predictive behavioir, specially by local changes in power features that would change the explanability rule between discovery algorithms.
PC-LiNGAM





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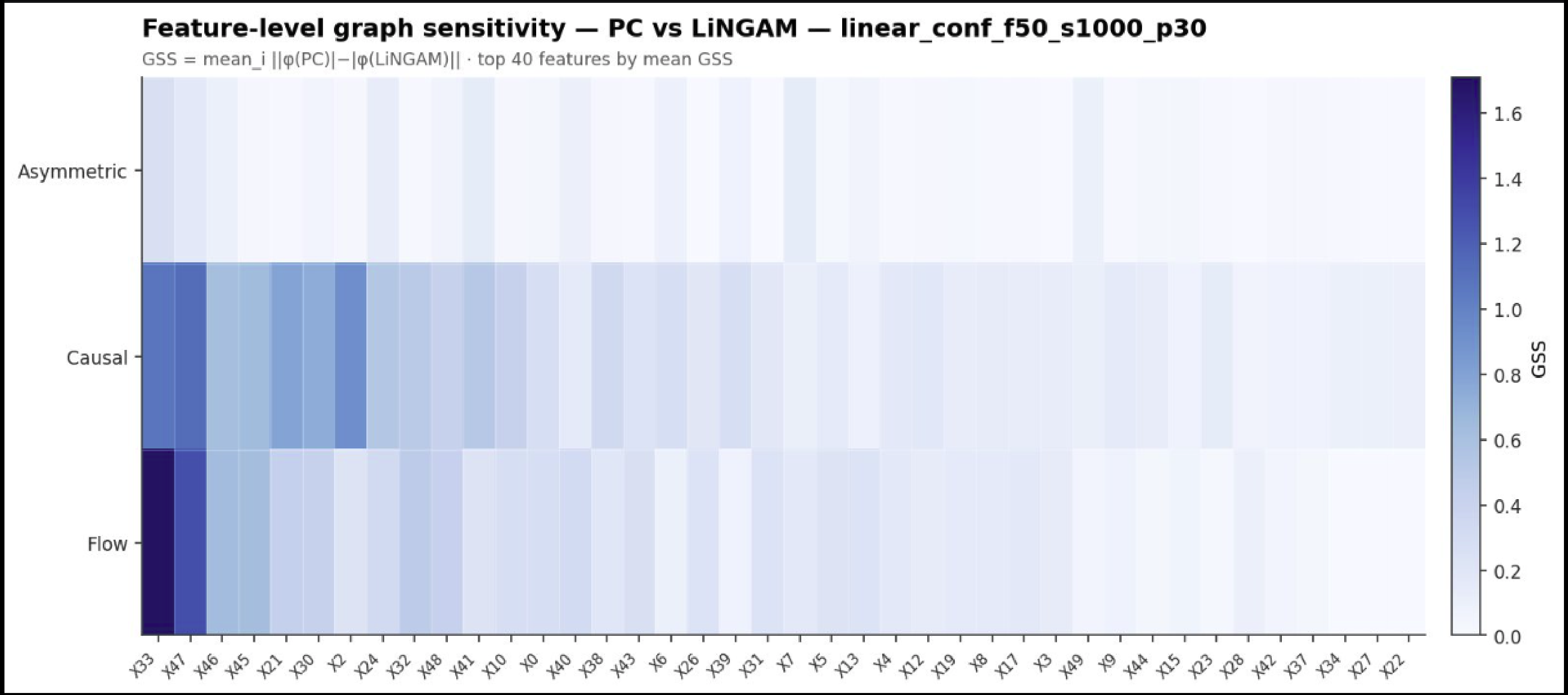

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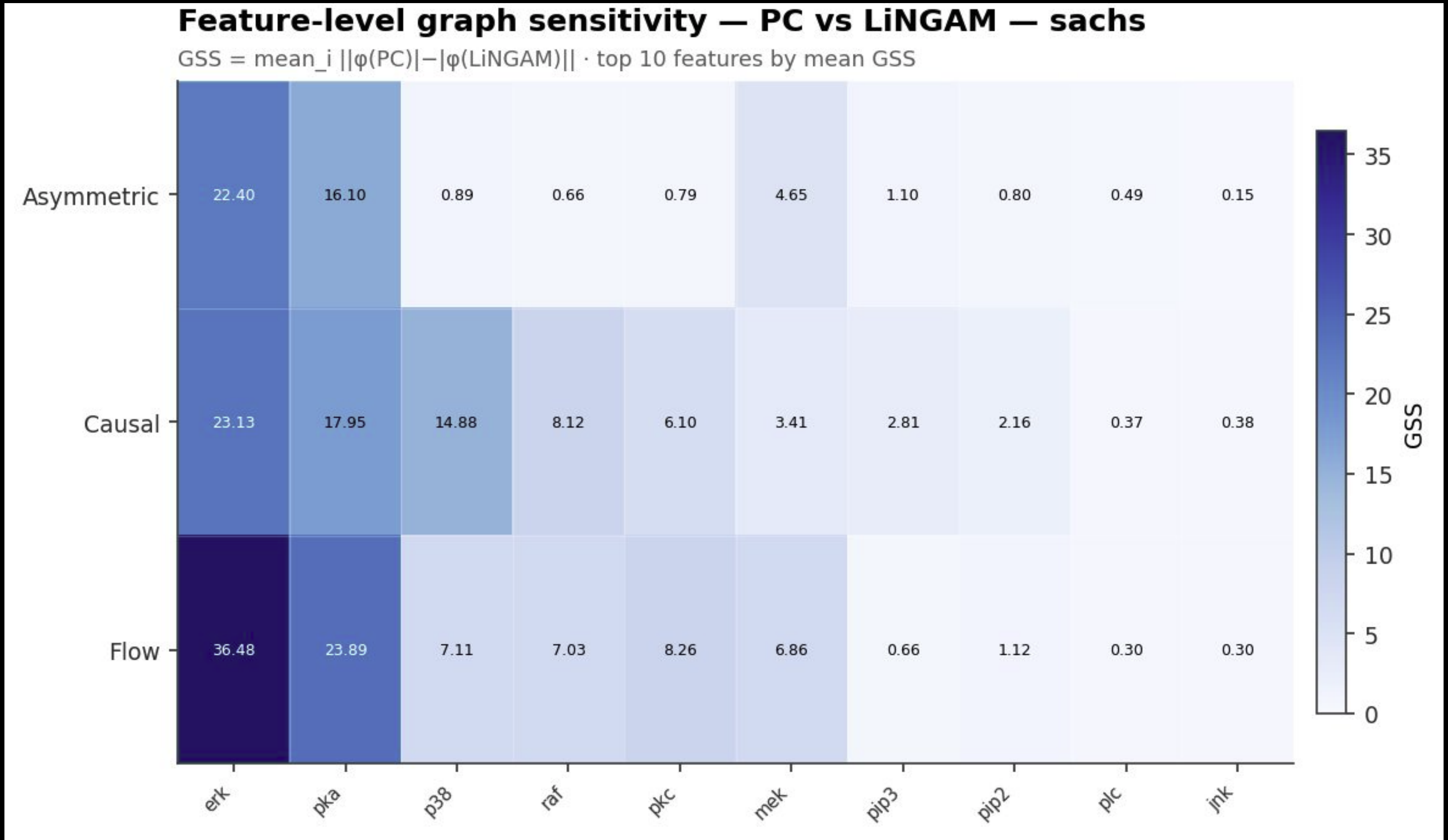
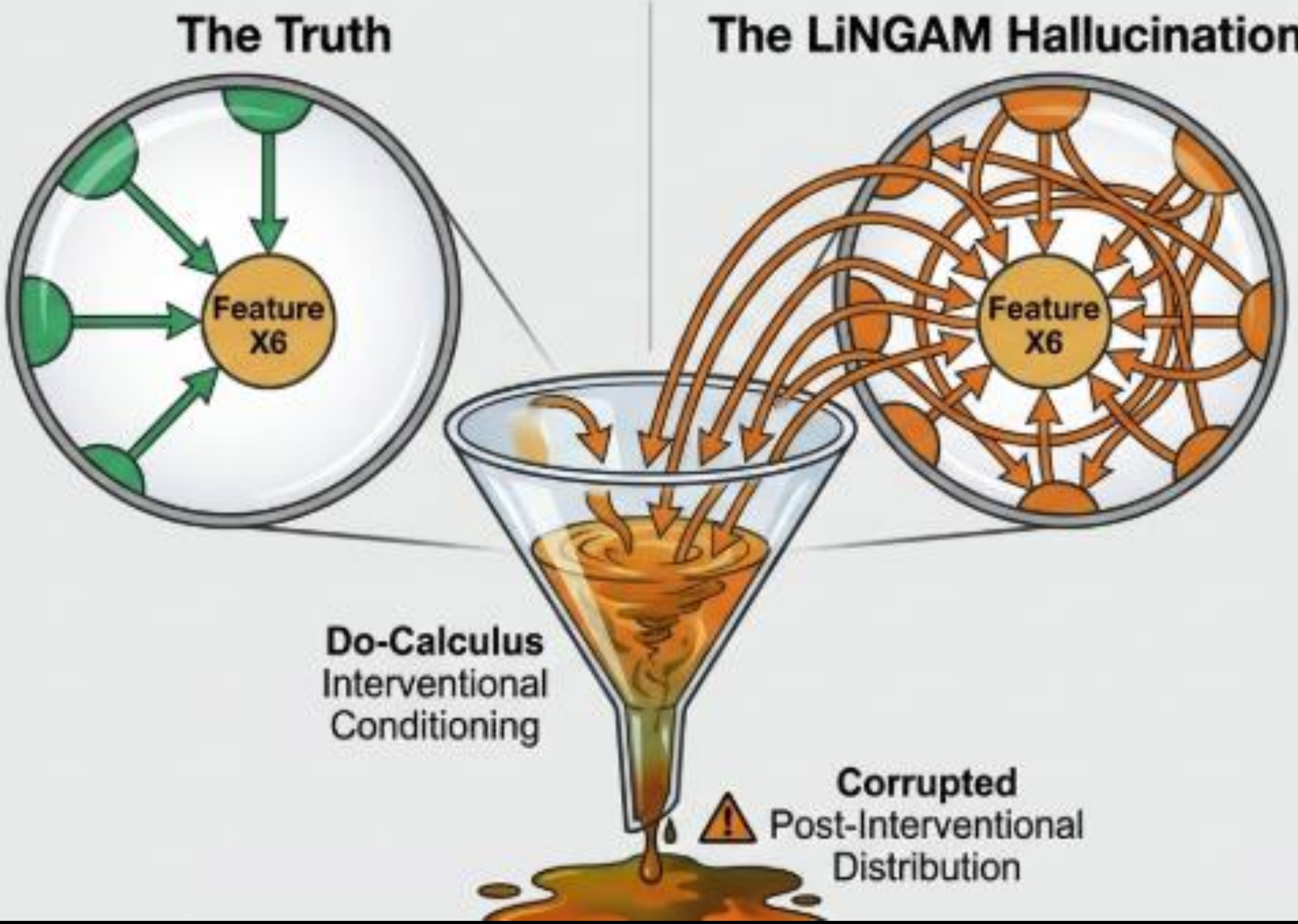


Anatomy of an Error: Interventional Contamination

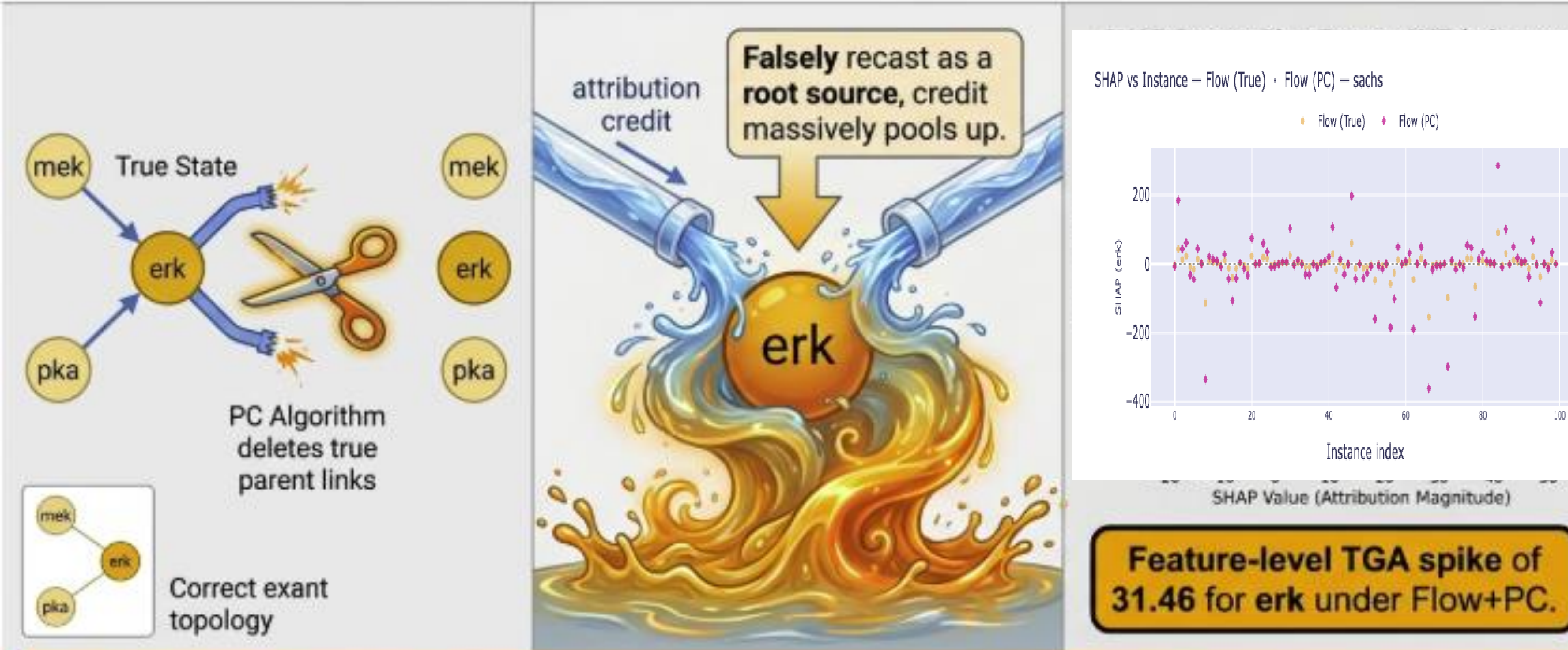


The Victim: Feature X6.

The Mechanism: Causal Shapley (CSV) uses incoming edges to perform interventional conditioning.

The Failure: When LiNGAM hallucinates 5 fake parent edges, CSV is forced to compute post-interventional distributions that do not correspond to any real causal mechanism. The discovery failure directly becomes an explainability failure.

Anatomy of an Error: Edge-Routing Chaos

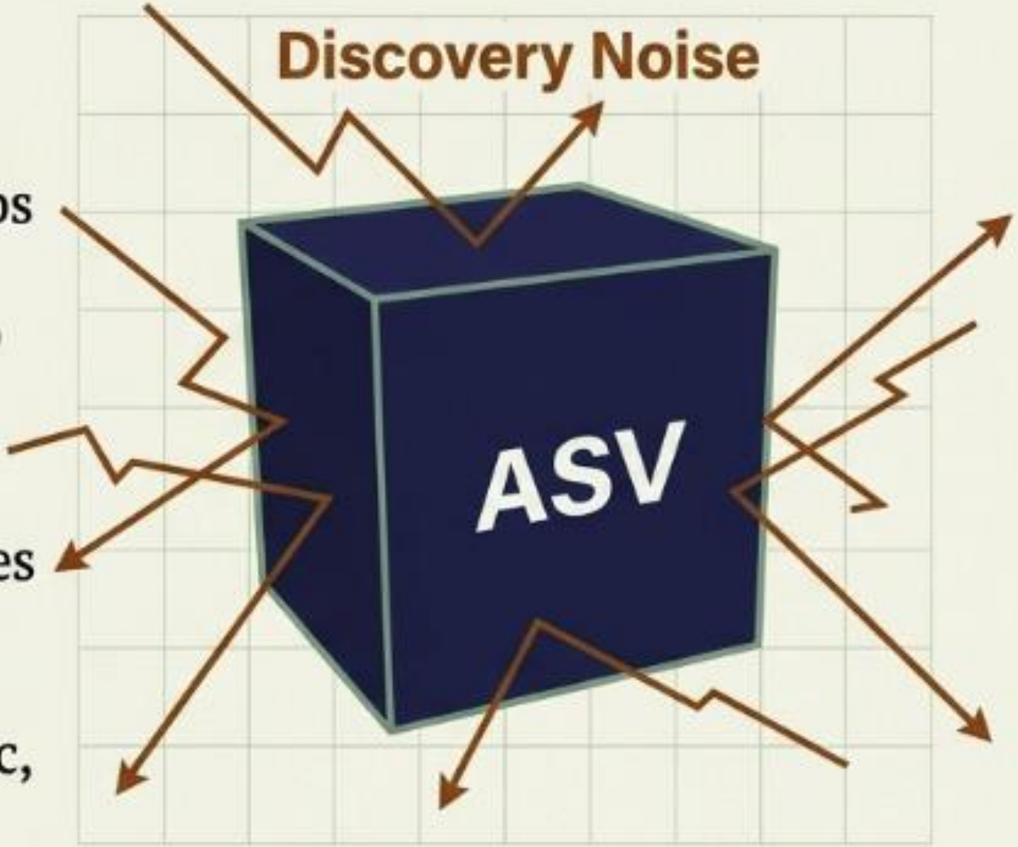


The Failure: Shapley Flow routes attribution strictly along edges. When algorithms delete or reverse links, downstream nodes accumulate massive artificial credit.

Asymmetric Shapley Values (ASV): The Safe Default

1. The Sparsity Buffer: ASV only restricts the permutation space. It absorbs discovery errors rather than directly inheriting them into the value function.

2. Oracle Fidelity: Achieves near-perfect alignment with ground-truth DAGs (<1% magnitude error on synthetic, 2.84% on Sachs).



3. Low Baseline Deviation: Rarely flips signs arbitrarily.

4. Zero Extra Cost: Adds no computational overhead beyond standard Monte Carlo SHAP.

When structural uncertainty is high, Causal Shapley and Flow are dangerous gambles. ASV is the safest default.

The Diagnostic Playbook for ML Teams

1. Baseline Check

Run Traditional Shapley + your chosen Method/Graph.

Measure Baseline Magnitude and Sign.

Flag features shifting massively.

2. Cross-Discovery Check

Run a second discovery algorithm (e.g., PC vs LiNGAM).

Measure Cross-Discovery Magnitude and Sign.

Flag contested neighborhoods.

3. The Two-Signal Audit

Overlay the results.

- High Magnitude = Validate dominant node edges.
- High Sign = Validate near-zero feature stability.

4. Domain Resolution

Use human expert knowledge to manually resolve the flagged, contested edges.

Only then, trust the explanation.